Cyber Security Task 3 – Secure File Sharing System

Overview

This task demonstrates a secure file sharing system where uploaded files are protected using AES encryption. Files are encrypted before storage and decrypted only when downloaded, ensuring data confidentiality.

Technology Used

- Backend: Python Flask: https://flask.palletsprojects.com/en/latest/
- Encryption: AES (PyCryptodome): https://pycryptodome.readthedocs.io/en/latest/
- Frontend: HTML / CSS
- Version Control: Git & GitHub

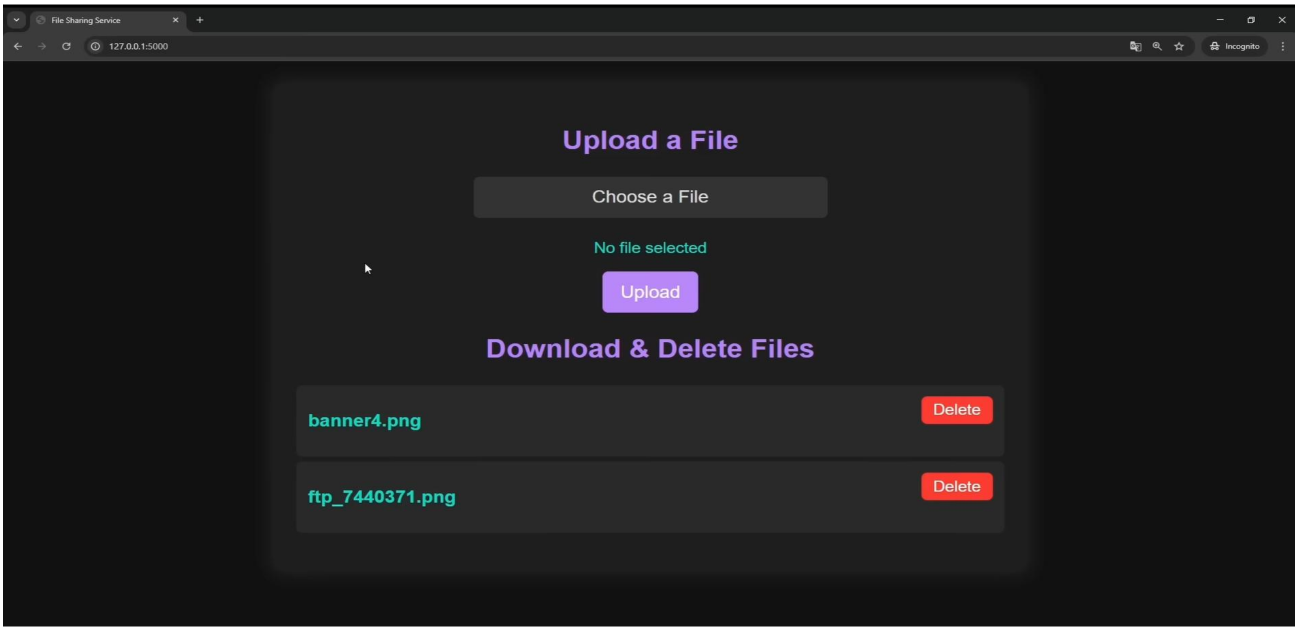
System Workflow

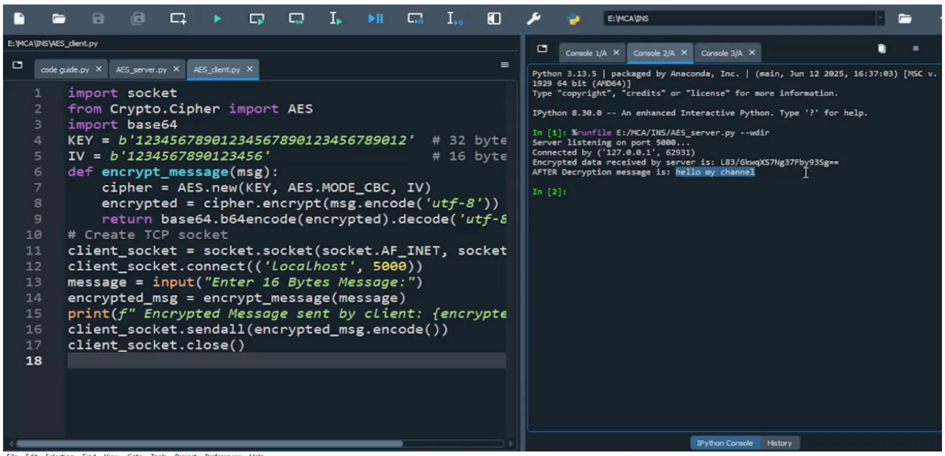
- User uploads a file using the web interface
- File is encrypted using AES
- Encrypted file is stored securely on the server
- On download request, the file is decrypted and sent to the user

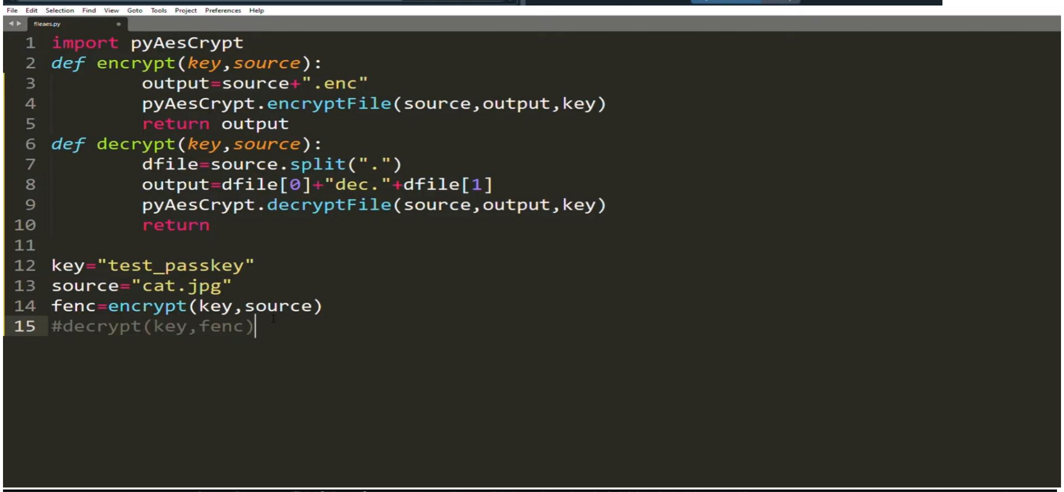
Key Security Features

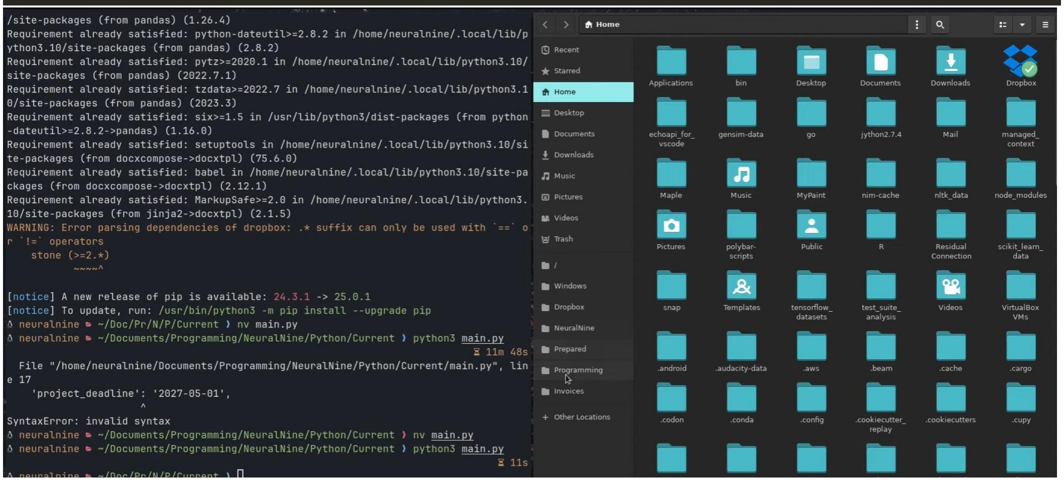
- AES encryption for files at rest
- Secure handling of encryption keys
- File integrity maintained during upload and download
- Restricted access to encrypted files

Screenshots









Security Overview

- AES (Advanced Encryption Standard) is used to protect files
- Encryption occurs before saving the file
- Decryption occurs only during file download
- Encryption keys are securely managed on the server

Testing Performed

- Uploaded sample files (PDF, TXT)
- Verified files are stored in encrypted format
- Confirmed successful decryption during download
- Ensured original file content remains unchanged

Conclusion

This project demonstrates basic secure file sharing techniques, encryption implementation, and safe file handling practices used in real-world systems.